Convergence of Gradient Descent and SGD

Your Name

Optimization setup

We minimize a differentiable objective, often written as an empirical average:

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}), \qquad f(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{x}), \qquad \mathbf{x}^\star \in \arg\min_{\mathbf{x}} f(\mathbf{x}).$$

Gradient descent uses the full gradient; stochastic gradient descent uses an estimate:

$$\text{GD:} \quad \mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \nabla f(\mathbf{x}_t),$$

$$\text{SGD:} \quad \mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \mathbf{g}_t, \qquad \mathbb{E}[\mathbf{g}_t \mid \mathbf{x}_t] = \nabla f(\mathbf{x}_t).$$

Goal

Choose the step sizes η_t and iteration budget T so that optimization error is provably small.

Geometry of the objective

L-smoothness controls how quickly the gradient can change:

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq L\|\mathbf{x} - \mathbf{y}\|_2.$$

μ-strong convexity gives the objective uniform curvature:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|_2^2.$$

The ratio $\kappa = L/\mu \geq 1$ is the condition number. Large κ means that different directions have very different curvature.

Hierarchy

Strong convexity gives geometric convergence; convexity alone generally gives a sublinear rate.

One gradient step decreases the objective

Let $\mathbf{x}^+ = \mathbf{x} - \eta \nabla f(\mathbf{x})$ be one gradient-descent step.

Lemma 1

If f is L-smooth, then

$$f(\mathbf{x}^+) \le f(\mathbf{x}) - \eta\left(1 - \frac{L\eta}{2}\right)\|\nabla f(\mathbf{x})\|_2^2.$$

In particular, every step with $0 < \eta \le 1/L$ satisfies

$$f(\mathbf{x}^+) \le f(\mathbf{x}) - \frac{\eta}{2}\|\nabla f(\mathbf{x})\|_2^2.$$

Meaning

Smoothness turns the gradient norm into a certified amount of progress.

Gradient descent for convex objectives

Assume that f is convex and L-smooth, and use a constant step size $0 < \eta \leq 1/L$.

Theorem 1

For every $T \geq 1$, gradient descent satisfies

$$f(\mathbf{x}_T) - f(\mathbf{x}^\star) \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^\star\|_2^2}{2\eta T}.$$

With $\eta = 1/L$ and $R = \|\mathbf{x}_0 - \mathbf{x}^\star\|_2$, it is enough to take

$$T \geq \frac{LR^2}{2\varepsilon} \qquad \Longrightarrow \qquad f(\mathbf{x}_T) - f(\mathbf{x}^\star) \leq \varepsilon.$$

Rate

Smooth convex optimization gives $f(\mathbf{x}_T) - f^\star = \mathcal{O}(1/T)$.

Strong convexity gives a linear rate

Now suppose that f is also μ-strongly convex and set $\eta = 1/L$.

Theorem 2

Gradient descent contracts the objective gap geometrically:

$$f(\mathbf{x}_T) - f(\mathbf{x}^\star) \le \left(1 - \frac{\mu}{L}\right)^T \big(f(\mathbf{x}_0) - f(\mathbf{x}^\star)\big) \le e^{-T/\kappa}\big(f(\mathbf{x}_0) - f^\star\big).$$

Well conditioned.
When $\kappa \approx 1$, a small number of full-gradient steps is enough.

Ill conditioned.
When $\kappa \gg 1$, progress is slow along low-curvature directions.

Complexity

Reaching error ε takes $\mathcal{O}(\kappa \log(1/\varepsilon))$ iterations.

Stochastic gradients

SGD replaces the full gradient by a conditionally unbiased estimator:

$$\mathbb{E}[\mathbf{g}_t \mid \mathbf{x}_t] = \nabla f(\mathbf{x}_t), \qquad \mathbb{E}\big[\|\mathbf{g}_t - \nabla f(\mathbf{x}_t)\|_2^2 \mid \mathbf{x}_t\big] \le \sigma^2.$$

The update has a deterministic component and a zero-mean noise component:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \nabla f(\mathbf{x}_t) - \eta_t \underbrace{\big(\mathbf{g}_t - \nabla f(\mathbf{x}_t)\big)}_{\text{gradient noise}}.$$

A mini-batch of size b reduces the variance of an average of independent gradients:

$$\mathbf{g}_t = \frac{1}{b}\sum_{j=1}^{b} \nabla f_{i_{t,j}}(\mathbf{x}_t), \qquad \mathbb{E}\|\mathbf{g}_t - \nabla f(\mathbf{x}_t)\|_2^2 \le \frac{\sigma^2}{b}.$$

Trade-off

Small batches make cheap progress; large batches produce a more accurate direction.

The fundamental SGD recursion

Assume that f is L-smooth and μ-strongly convex. Define

$$a_t = \mathbb{E}\|\mathbf{x}_t - \mathbf{x}^\star\|_2^2, \qquad c = \frac{2\mu L}{\mu + L}.$$

Proposition 1

For an unbiased stochastic gradient with variance at most σ^2, every step size $0 < \eta_t \leq 2/(\mu + L)$ satisfies

$$a_{t+1} \leq (1 - c\eta_t)a_t + \eta_t^2\sigma^2.$$

The two terms have different roles:

$$\underbrace{(1 - c\eta_t)a_t}_{\text{contraction toward } \mathbf{x}^\star} + \underbrace{\eta_t^2\sigma^2}_{\text{error injected by noise}}.$$

Consequence

A useful schedule must preserve contraction while eventually suppressing the noise.

Diminishing steps converge to the optimum

Use the recursion from Proposition 1 and choose

$$\eta_t = \frac{2}{c(t+t_0)}, \qquad t_0 \geq \frac{\mu + L}{c}, \qquad c = \frac{2\mu L}{\mu + L}.$$

Theorem 3

Let

$$C = \max\left\{ t_0 a_0, \frac{4\sigma^2}{c^2} \right\}.$$

Then the expected squared distance obeys

$$\mathbb{E}\|\mathbf{x}_T - \mathbf{x}^\star\|_2^2 = a_T \leq \frac{C}{T + t_0}.$$

The offset t_0 keeps every early step inside the stable range. The $1/t$ decay balances contraction against stochastic noise.

Rate

Diminishing-step SGD reaches the exact optimum at the sublinear rate $\mathcal{O}(1/T)$ in squared distance.

Constant steps reach a noise floor

Fix $0 < \eta \leq 2/(\mu + L)$. Iterating the fundamental recursion gives

$$a_T \leq (1 - c\eta)^T a_0 + \eta^2 \sigma^2 \sum_{k=0}^{T-1} (1 - c\eta)^k.$$

Proposition 2

The geometric sum yields the explicit bound

$$a_T \leq (1 - c\eta)^T a_0 + \frac{\eta \sigma^2}{c}.$$

Consequently,

$$\limsup_{T \to \infty} \mathbb{E} \|\mathbf{x}_T - \mathbf{x}^\star\|_2^2 \leq \frac{\eta \sigma^2}{c}.$$

Consequence

A larger constant step converges faster at first, but settles in a wider neighborhood of the optimum.

Appendix

Assumptions at a glance

The deterministic and stochastic analyses use the same objective geometry:

$$\mu \mathbf{I} \preceq \nabla^2 f(\mathbf{x}) \preceq L \mathbf{I} \qquad \text{when the Hessian exists.}$$

Equivalent tools

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq L \|\mathbf{x} - \mathbf{y}\|_2,$$

$$\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq \frac{\mu L}{\mu + L} \|\mathbf{x} - \mathbf{y}\|_2^2 + \frac{1}{\mu + L} \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2^2.$$

- L controls the largest stable step; μ controls contraction.
- σ^2 measures stochastic-gradient variance; b is the mini-batch size.
- $\kappa = L/\mu$ summarizes conditioning.

Proof of the descent lemma

Smoothness gives a quadratic upper model around $\mathbf{x}$:

$$f(\mathbf{y}) \le f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|_2^2.$$

Substitute the update

Set $\mathbf{y} = \mathbf{x}^+ = \mathbf{x} - \eta \nabla f(\mathbf{x})$. Then

$$\langle \nabla f(\mathbf{x}), \mathbf{x}^+ - \mathbf{x} \rangle = -\eta \|\nabla f(\mathbf{x})\|_2^2,$$

$$\|\mathbf{x}^+ - \mathbf{x}\|_2^2 = \eta^2 \|\nabla f(\mathbf{x})\|_2^2.$$

Collecting both terms produces the exact one-step decrease.

Result

$$f(\mathbf{x}^+) \le f(\mathbf{x}) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(\mathbf{x})\|_2^2.$$

Proof of linear convergence

Strong convexity implies the Polyak–Łojasiewicz inequality

$$\|\nabla f(\mathbf{x})\|_2^2 \geq 2\mu\big(f(\mathbf{x}) - f(\mathbf{x}^\star)\big).$$

Combine

With $\eta = 1/L$, the descent lemma gives $f(\mathbf{x}_{t+1}) - f^\star \leq (1 - \mu/L)\big(f(\mathbf{x}_t) - f^\star\big)$.

Iterate

Applying the contraction T times yields $f(\mathbf{x}_T) - f^\star \leq (1 - 1/\kappa)^T\big(f(\mathbf{x}_0) - f^\star\big)$.

What mini-batching changes

Average b independent stochastic gradients at the same iterate:

$$\mathbf{g}_t^{(b)} = \frac{1}{b}\sum_{j=1}^{b} \mathbf{g}_{t,j}, \qquad \mathbb{E}\|\mathbf{g}_t^{(b)} - \nabla f(\mathbf{x}_t)\|_2^2 \le \frac{\sigma^2}{b}.$$

The mean update direction is unchanged:

$$\mathbb{E}[\mathbf{g}_t^{(b)} \mid \mathbf{x}_t] = \nabla f(\mathbf{x}_t).$$

The fundamental recursion becomes

$$a_{t+1} \le (1 - c\eta_t)a_t + \frac{\eta_t^2 \sigma^2}{b}.$$

For a constant step size, the limiting term is

$$\limsup_{T\to\infty} a_T \le \frac{\eta\sigma^2}{bc}.$$

Practical message

Mini-batching lowers the noise floor by $1/b$, but each iteration uses b gradient evaluations. It is most valuable when those evaluations run efficiently in parallel.